

Programming In C

Difference between Java & C:

Object oriented concepts C is procedural not OO it does not support classes, objects, inheritance, or polymorphism.

Pointers. C has pointers while java does not

Memory Management. Java has the java garbage collector which automatically handles memory **garbage management** while in C we do that manually.

Execution environment C is machine code while java is byte code

Compilation Process is also different

Numerical Datatypes

String Datatypes

Pointer in C: A pointer is a variable that stores the memory address of another variable. Instead of holding a value directly, it “points” to the location in memory where the value is stored.

How do you get an address of a variable in C? Name the operator and its symbol: by simply printing the & + the variable. The & refers to the address. While * refers to the value

How are arguments of functions passed in C (like in Java)?: arguments are passed by value not by reference meaning that passing an argument inside a function only changes the value of this variable inside the function not its true value since what was passed was only a copy of this variable.

How do you call the fact, if an argument of a function could be changed within the function in C and the change is available outside of the function after its call? What is used for this in C-Language?: for this we use passing by reference which means we accept a pointer at the function definition and we should pass the value as a reference &int this way the function can change the original value of the variable after its ran

Linux commands:

top: table of processes this command prints list of all processes for all users on the system

ps: processes status, this command shows all the information about running processes on the system

ps -ef: this command lists all the processes for all users on the system.

Questions and Answers:

How do you start any shell command in the background?: when you write an & after the process name it automatically runs it into the background. Firefox & this command runs firefox as a process into the background

Which shell command can be used to stop a process?: we either press ctrl + z or we use kill - STOP PID command.

Which shell command can be used to terminate a process from the system?: kill + PID

Kill -9 PID terminates it immediately and remove it from the table of processes

What does the system command fork return?: for the child it returns 0, for the parent it returns an integer greater than 0, the parent returns the child PID during the fork. While it returns -1 in case there was any errors.

fork() creates a new child process as a copy of the parent and returns 0 to the child and the child's PID to the parent (or -1 on error). execl() replaces the current process image with a new program, keeping the same PID, and does not return if successful. wait() makes a parent process pause until a child process terminates, returning the child's PID and allowing the parent to collect its exit status. exit() terminates the current process and optionally returns a status code to the parent. **Comparison:** fork() creates a new process, execl() replaces a process, wait() pauses the parent until the child finishes, and exit() ends a process; often used together in parent-child process management.

Check how many and which processes are currently running in your environment. What command must be used for that?: solved using top or ps -ef

Stop the process of your text editor in the second shell using kill-s STOP . What process state does mousepad or gedit take now? What happens when you try to use the text editor?: for this one I used kill -STOP PID it froze the text editor and the text editor is not responding no matter how many times I click on it.

Set mousepad or gedit back to a running background process. What state information do you now get about this process? What happens when you enter text into the editor?: for this one I used bg + process number (1,2,3) and it is running back again in the background

What do you notice regarding the PIDs of the parent and child processes when executing the program? Which parent process is now indicated for the child process?: the PID of the child was different from that of the parent but the parent PID of the child was the same as the PID of the parent

When you change the maximum number of loop iterations, the parent and child may finish their loops at different times, so their outputs can interleave unpredictably. The child's parent process (PPID) still shows the original parent's PID.

A difficulty arises if parent and child need to exchange data because each process has its own separate memory space; changes in one process's variables (like counter) are not seen by the other, so they cannot directly share variables without using inter-process communication (IPC) mechanisms like pipes, shared memory, or files.

In the task 3.3, you may have already discovered the particular situation for orphaned child processes. a) Which C system command could you use to solve the issue? Research the command or commands that could prevent getting orphaned child processes: To prevent orphaned child processes, you can use the wait() system call in the parent process. This makes the parent wait for the child to terminate and collect its exit status, preventing the child from becoming orphaned. Alternatively, you can use waitpid() for more control over which child to wait for.

```
int main() {  
  
    pid_t pid = fork();  
  
    if (pid < 0) {
```

```
    perror("fork failed");  
    exit(EXIT_FAILURE);  
}  
  
if (pid == 0) { // Child  
    execl("/bin/ls", "ls", "-l", "/home/student", NULL);  
    perror("execl failed");  
    exit(EXIT_FAILURE);  
} else { // Parent  
    wait(NULL); // Wait for child  
    printf("Child finished listing files.\n");  
}  
  
return 0;  
}
```